Summary: Social Media Engagement Analytics

This summary provides a comprehensive overview of the social media engagement dataset of 5,000 records

1. Data Understanding and Cleaning

- **Data Structure:** The raw data has exactly **5,000 rows** and **20 distinct columns**.
- **Attributes:** user_id, age, gender, country, post_id, post_type, post_category, likes, comments, shares, watch_time_sec, impression_count, posted_at, follower_count, is_verified, device_type, sentiment, hashtags, engagement_rate, and hashtag_count.
- **Missing Value Imputation:** Missing values identified in age, likes, shares, and comments (150 missing records each) were resolved using the first appearing mode. Categorical null values in gender and sentiment were grouped under an "Unknown" label. No duplicate records were found.
- **Feature Engineering:**
    - posted_at was successfully standardized from text strings to datetime objects.
    - hashtag_count was extracted, revealing a maximum user usage of 3 hashtags per post.
    - A new metric, engagement_score (sum of likes, comments, and shares), was introduced to measure raw, volume-based audience activity.

2. Baseline Descriptive Statistics

- **Age:** Spans from 13 to 64 years, with an overall average user age of 38.2 years.
- **Likes:** Represents the dominant driver of audience interaction, averaging 9,998.41 likes per post (ranging from 10 to 19,998).
- **Comments & Shares:** Average 1,545.15 comments and 976.69 shares per post.
- **Watch Time & Impressions:** The average video/reel watch time sits at 4,014.5 seconds (~67 minutes), generating a mean impression count of 50,013.73 per post.

3. Key Analytics & Performance Insights

Demographics & Behavioral Trends

- **Age Fluctuation:** The relationship between age and the average engagement rate is non-linear and volatile. It features a sharp, primary peak around **ages 24–25**, followed by a prominent secondary middle-age surge between **ages 51 and 55**.
- **Weekday Efficiency: Tuesday** stands out as the single most effective day for maximizing reach, yielding the highest average impression count across the week.
- **Device Affinity:** The hardware type used (mobile, desktop ,tablet) has a negligible, near equal impact on consumption habits, though mobile users hold a marginal edge in watch time.

Content Performance

- **Format Superiority: Video** is the highest-performing content type, leading both in average likes and overall average engagement rate.
- **Category Dynamics: Food** contents secure the highest average engagement rate. Conversely, while Fashion and Education categories lead in average watch time, Fashion content generates the fewest average likes indicating that high watch-duration categories do not automatically convert into positive interactions.
- **Peak Engagement Event:** The maximum raw engagement score achieved in the dataset (24,740 interactions) was generated by a single **Lifestyle Video**.

Verification & Sentiment Factors

- **Verification Premium:** Verified accounts achieve a distinct performance advantage over unverified profiles, displaying a **10.4% higher average engagement rate**.
- **The Sentiment Paradox:** Although positive sentiment posts make up the highest volume in the dataset (2,363 posts), **negative sentiment posts yield a higher average engagement rate** (1.04) and more average likes (10,094.99). Neutral posts, however, prompt the highest volume of user discussion, leading slightly in average comment counts.